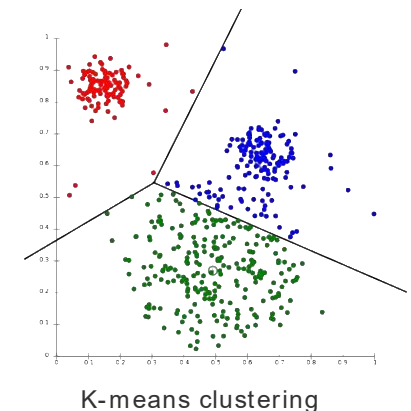
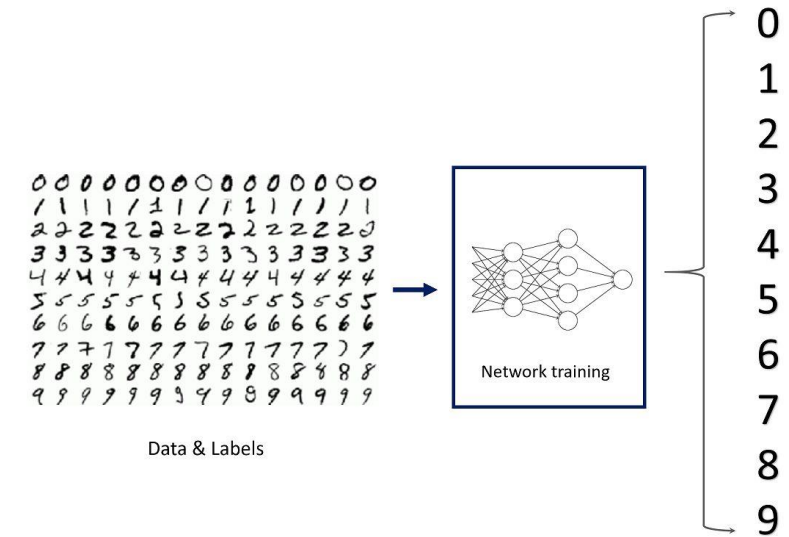


Generative Methods

—— GAN (Generative Adversarial Networks) ——

Supervised vs Unsupervised Learning

- Supervised - data + label
 - classification, regression, detection etc.
 - learning a function to map x to y
- Unsupervised - Just data + no labels
 - clustering, dimensionality reduction, feature learning etc.
 - Learn some underlying hidden structure of the data



Generative modeling

- Given training data, generate new samples from the same distribution



The 60 False-Positive pairs (1.00%) on LFW by DeepFace-ensemble.



Generative modeling

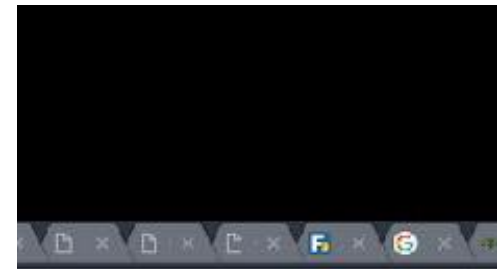
- Given training data, generate new samples from the same distribution



Chainer NewDeepLayer Eval NIPS2017

images and learn codes that contain
textures. However, GANs are
networks and it is important (and
example, they can oscillate between
tendency to collapse. In this work, Tim

Training data $\sim p_{\text{data}}(x)$ Generated samples $\sim p_{\text{model}}(x)$



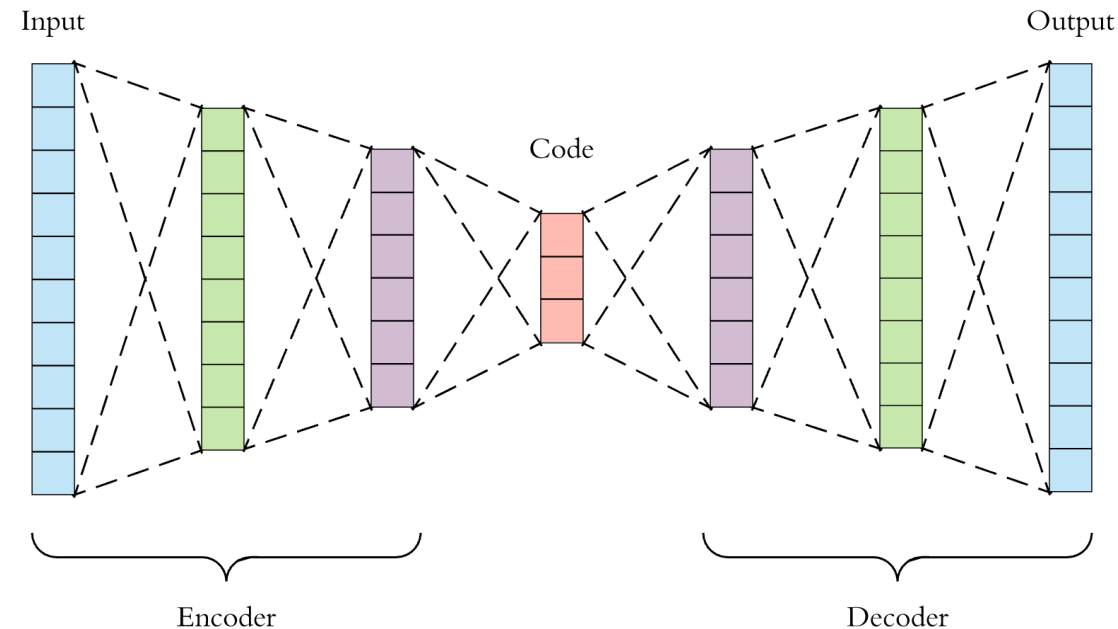
segNet Mathieu Weekend GAN

promising training on generated data
they produce very clean and sharp
valuable information about these
formulated as a game between two

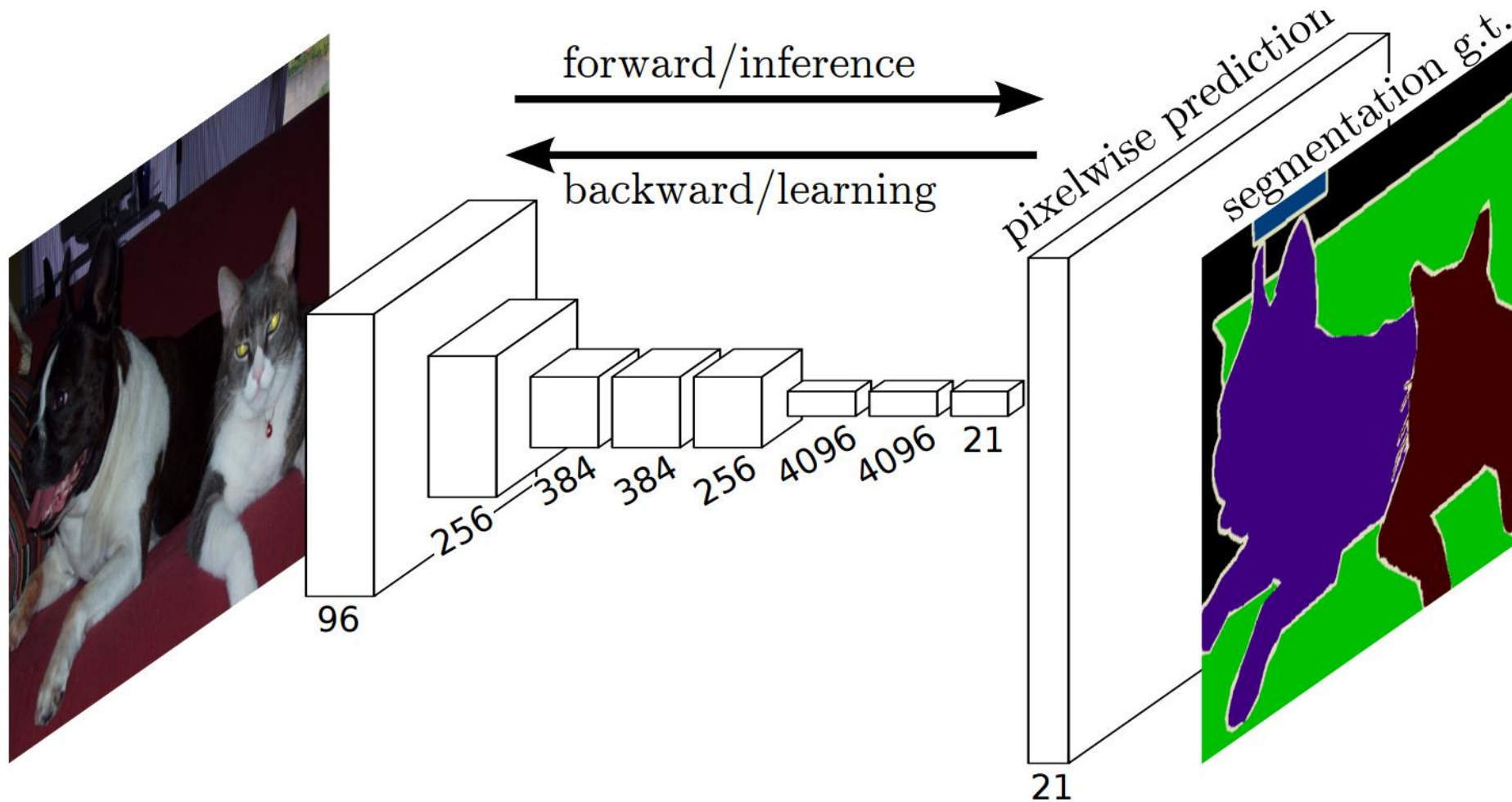
Want to: learn $p_{\text{model}}(x)$ similar to $p_{\text{data}}(x)$

Background - Auto encoders

- Encoder Decoder architecture
 - Use for compression
 - Or throw away the decoder after training use the compressed code for supervised learning tasks (particularly useful in problems with small training set)

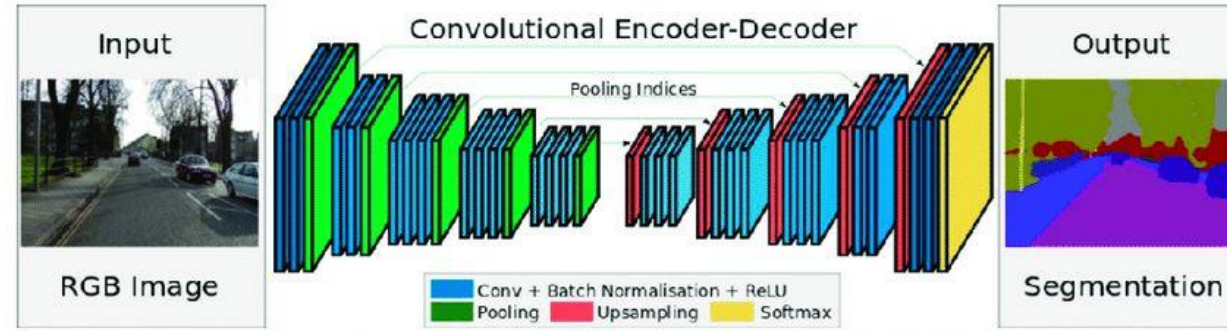


Background – Image to Image

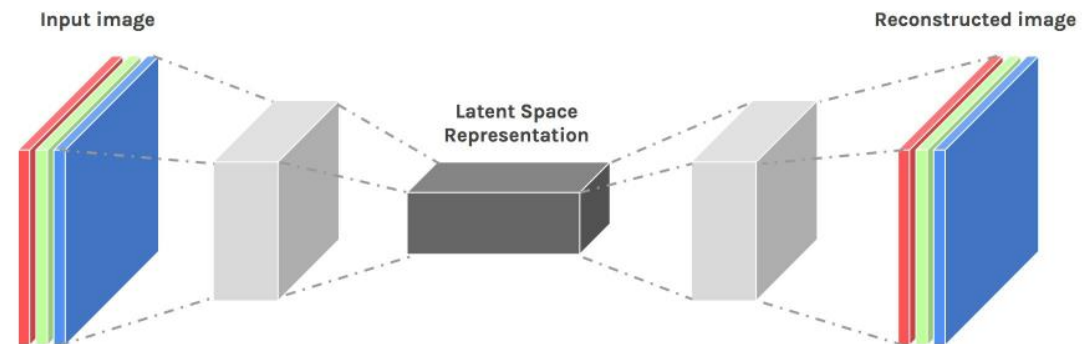


Generation

Semantic Segmentation



Neural Inpainting



Generative Models

Increasing Role of Synthetic Data

- Code to synthesis “Real data” ;
 - Role in data augmentation
 - Capability to generate for Neural Networks

Can we detect generated?

- The “story” of Police and “fake notes”

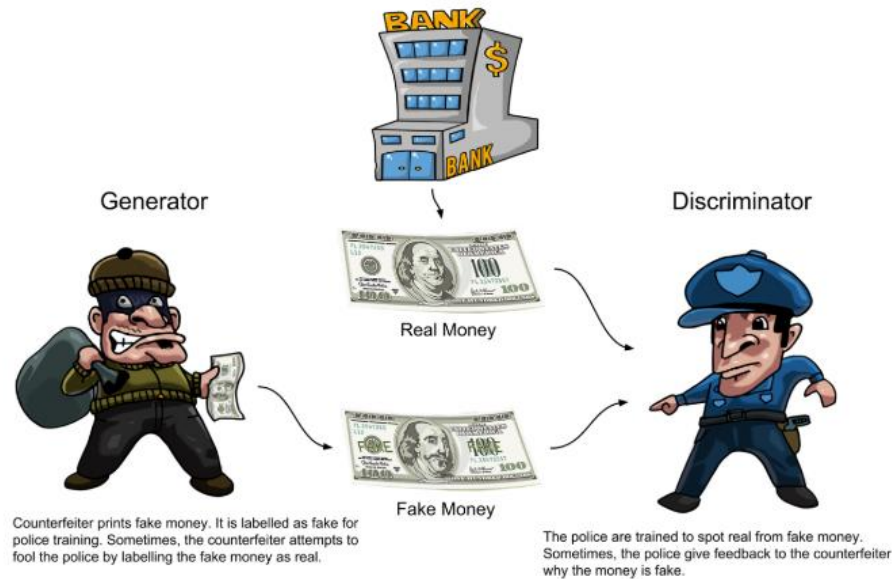
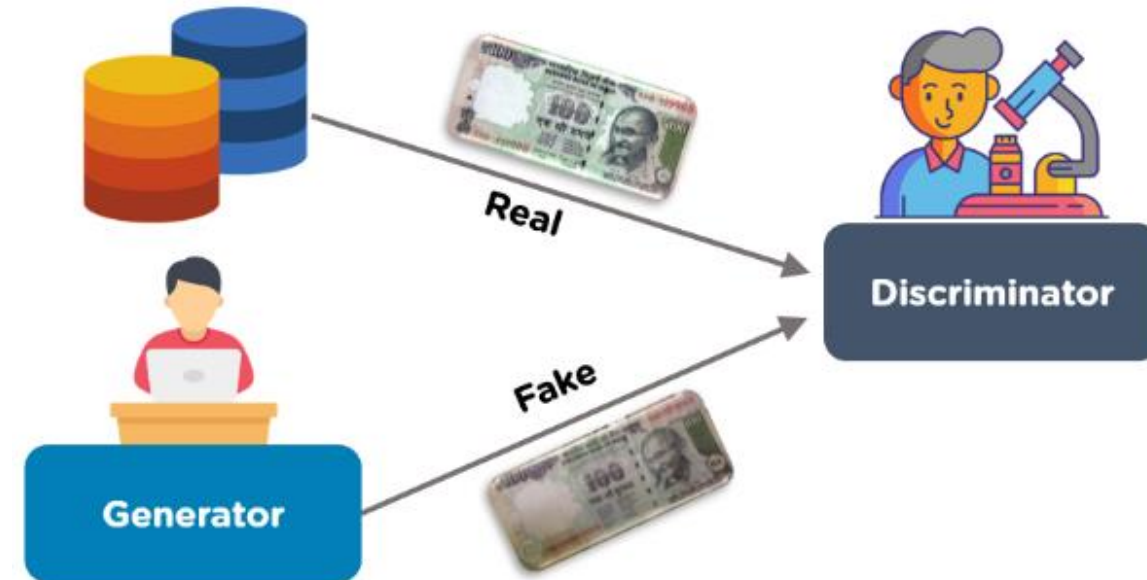
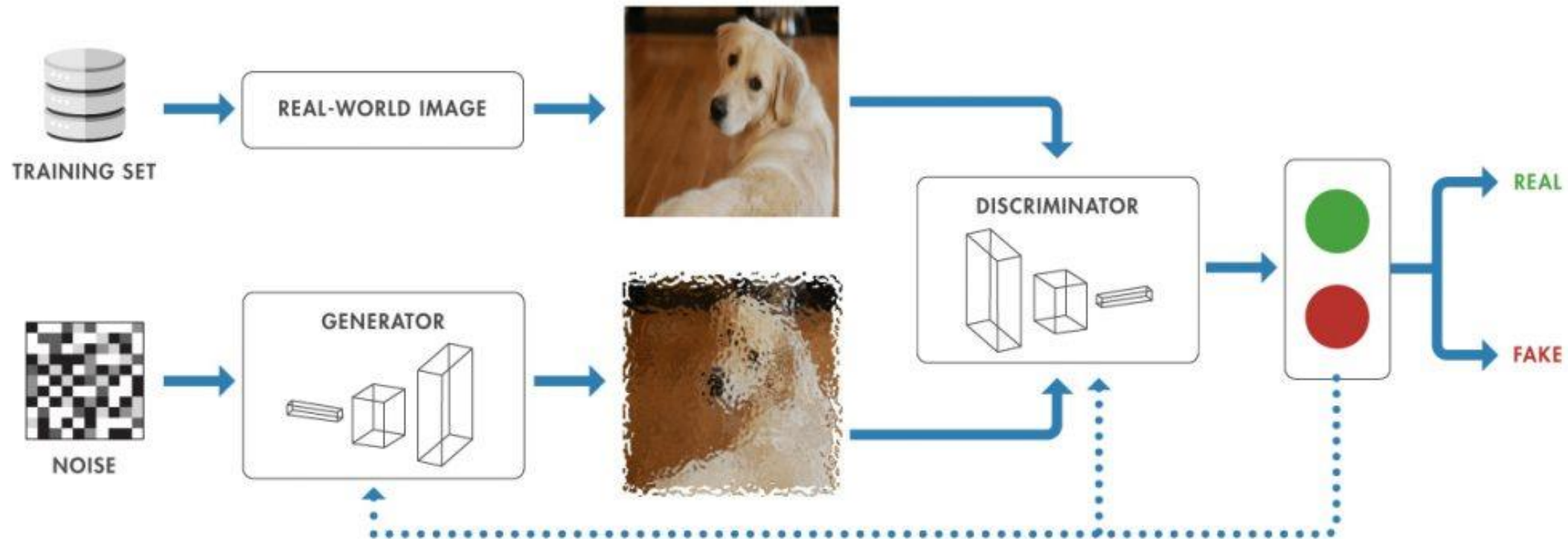


Image: Courtesy to Richard Gall.



Generation



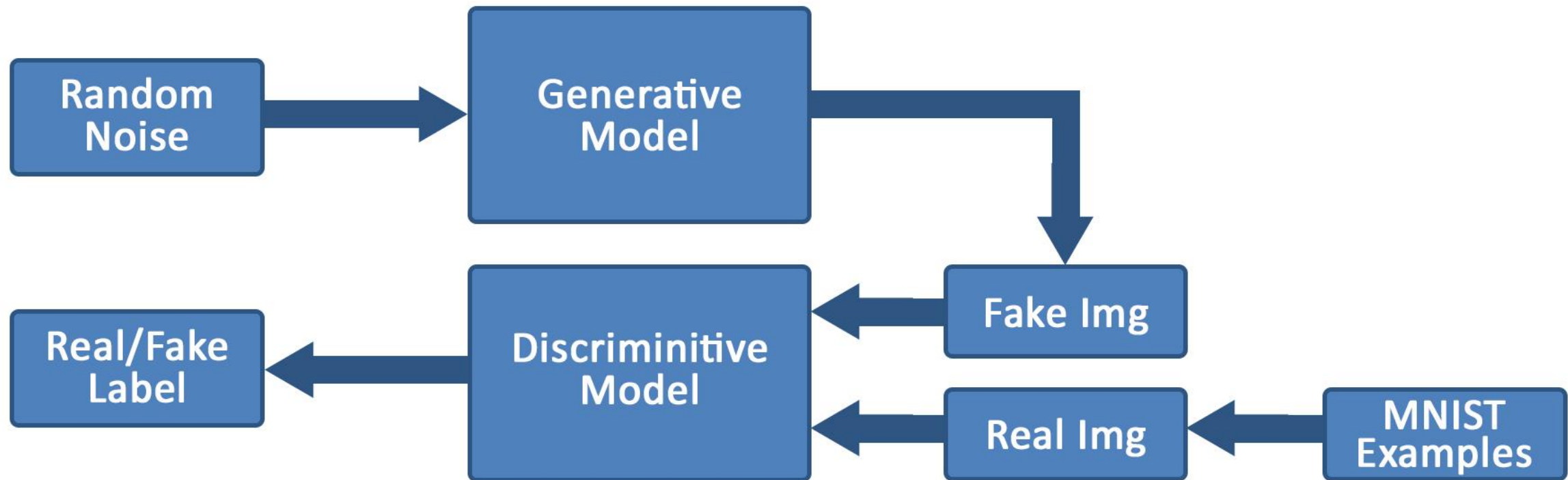
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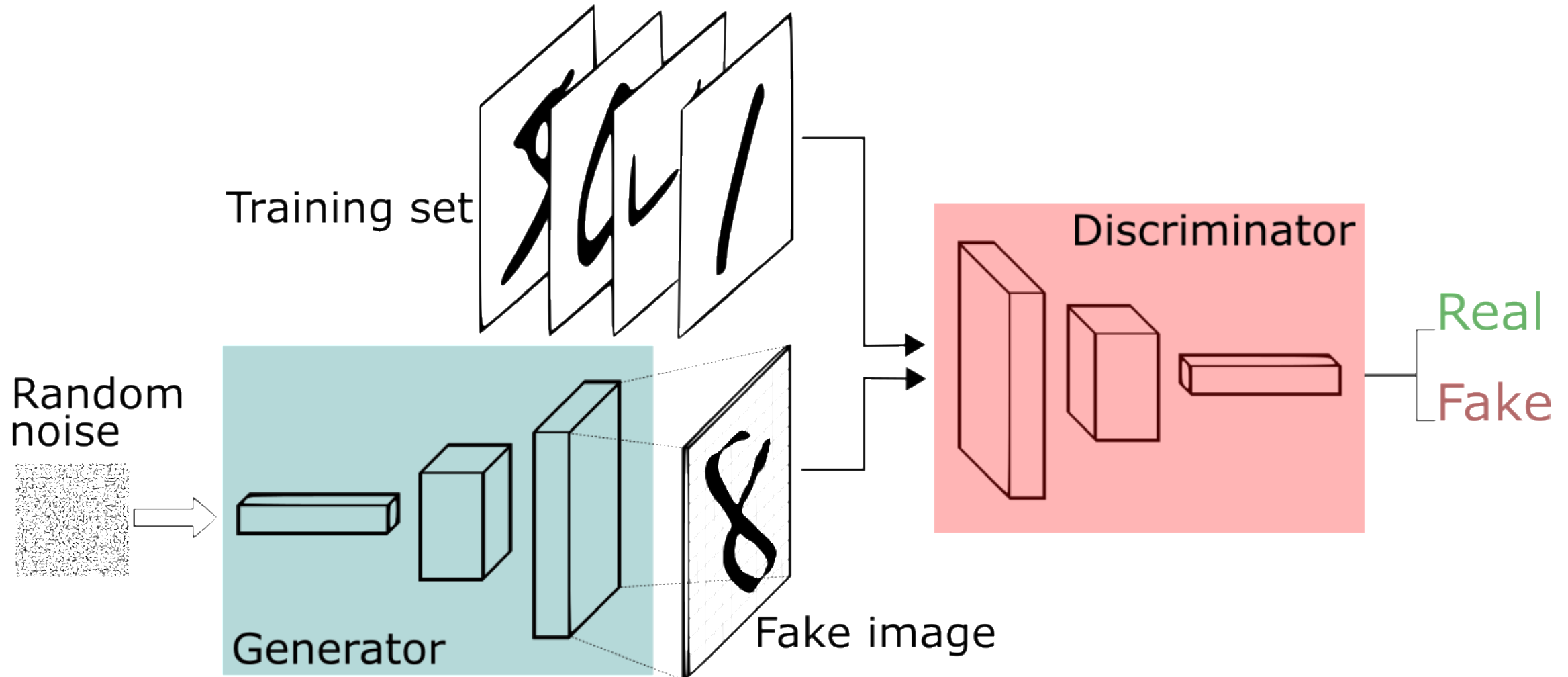
GANs

- Generative Adversarial Networks
- A game between two players:
 - Discriminator D
 - Generator G
- D tries to discriminate between:
 - A sample from the data distribution
 - A sample from the generator G
- G tries to “trick” D by generating samples that are hard for D to distinguish from data.

GAN: Network architecture

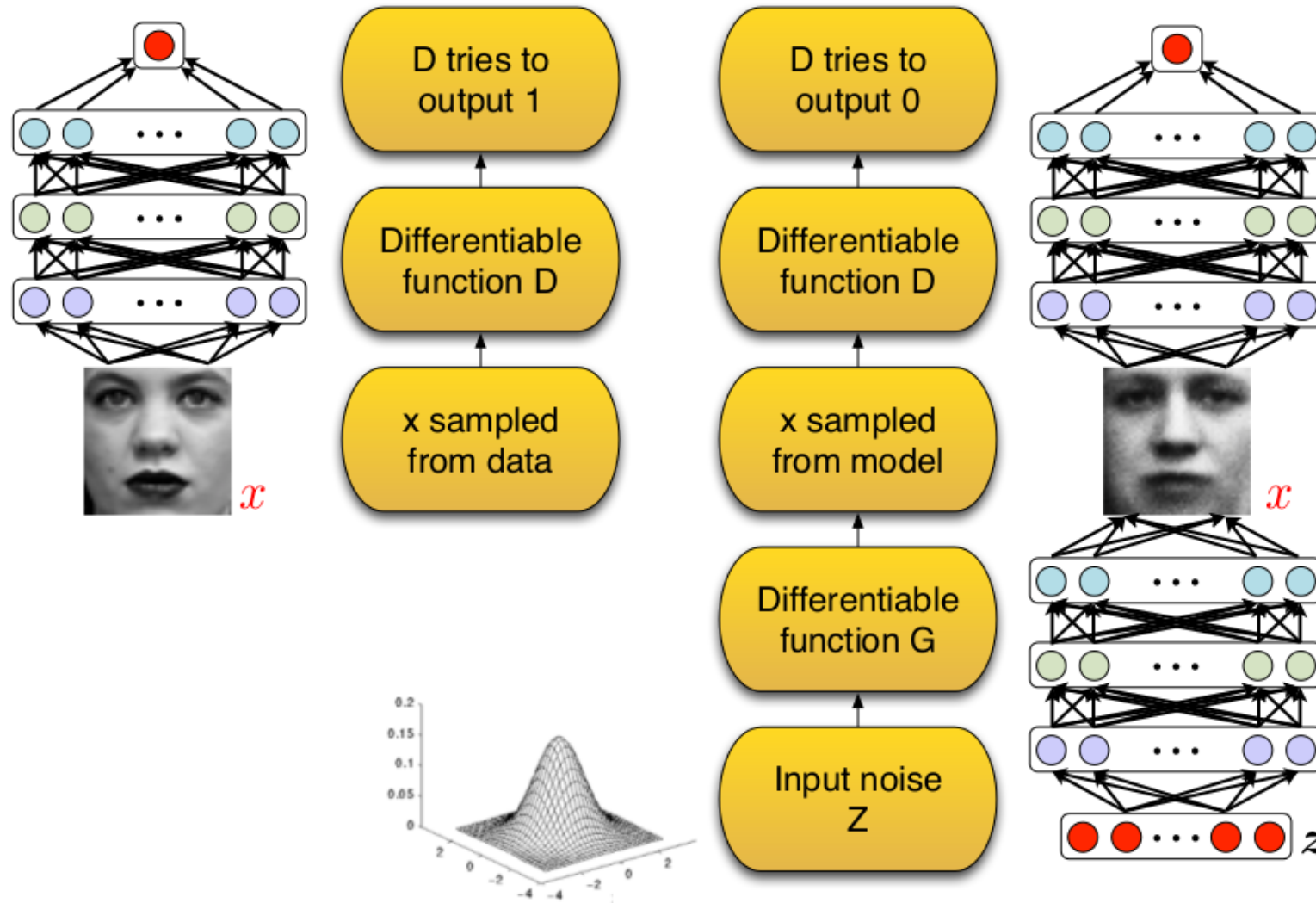


GAN: Network architecture

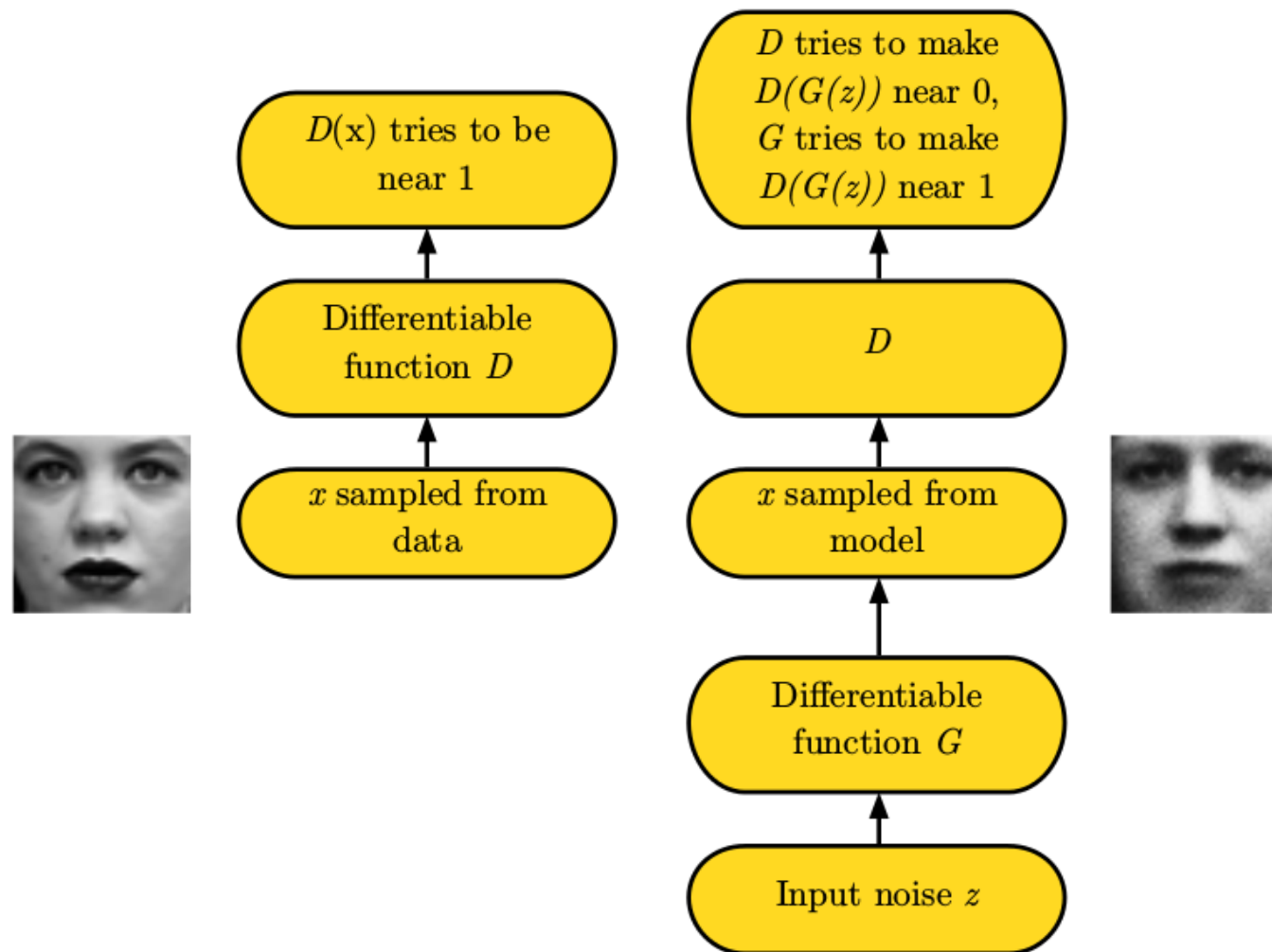


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GAN framework

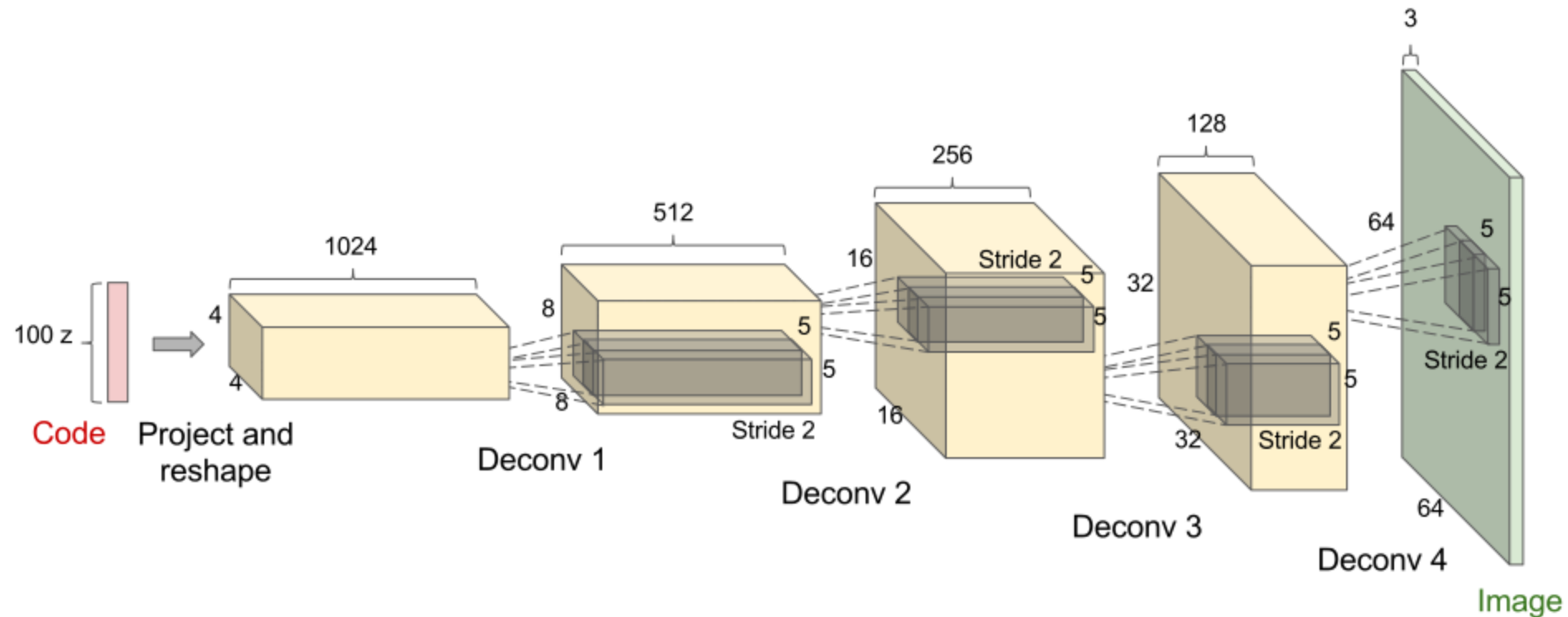


GANs

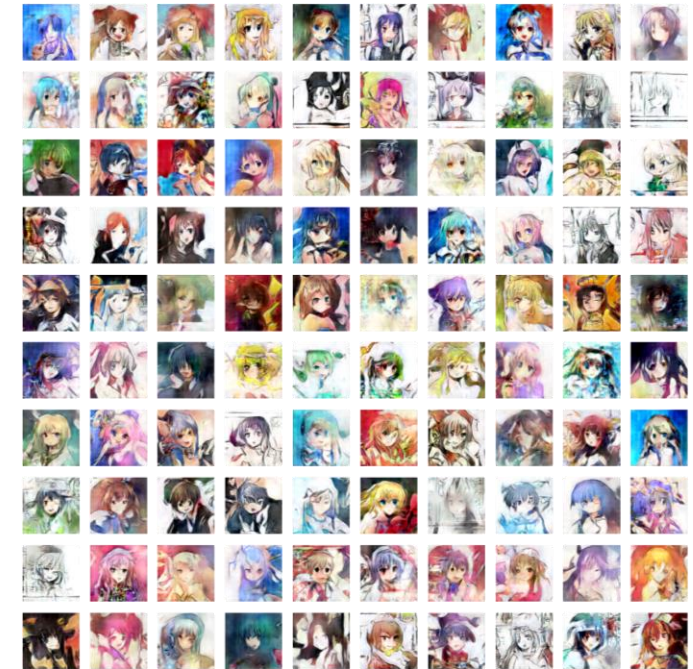


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Extension: Deep Convolutional GAN (DCGAN)



Samples (Synthetic Images)



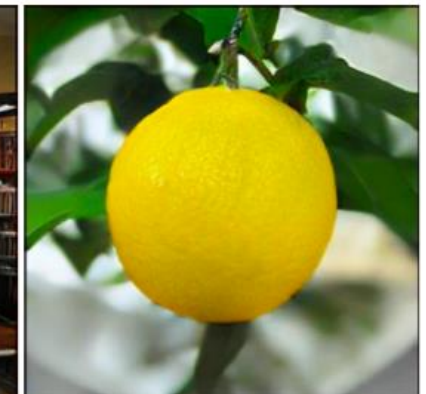
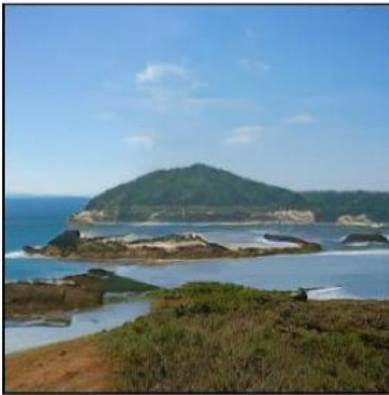
Results: Progressive GAN (ICLR 2018)



Results: Big GAN (ICLR 2019)

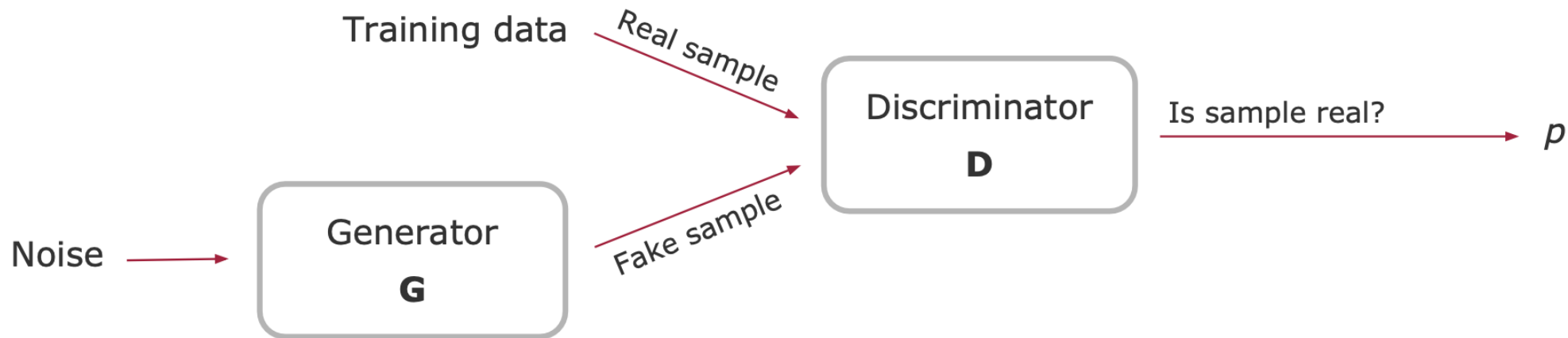


Results: Big GAN (ICLR 2019)



GANs

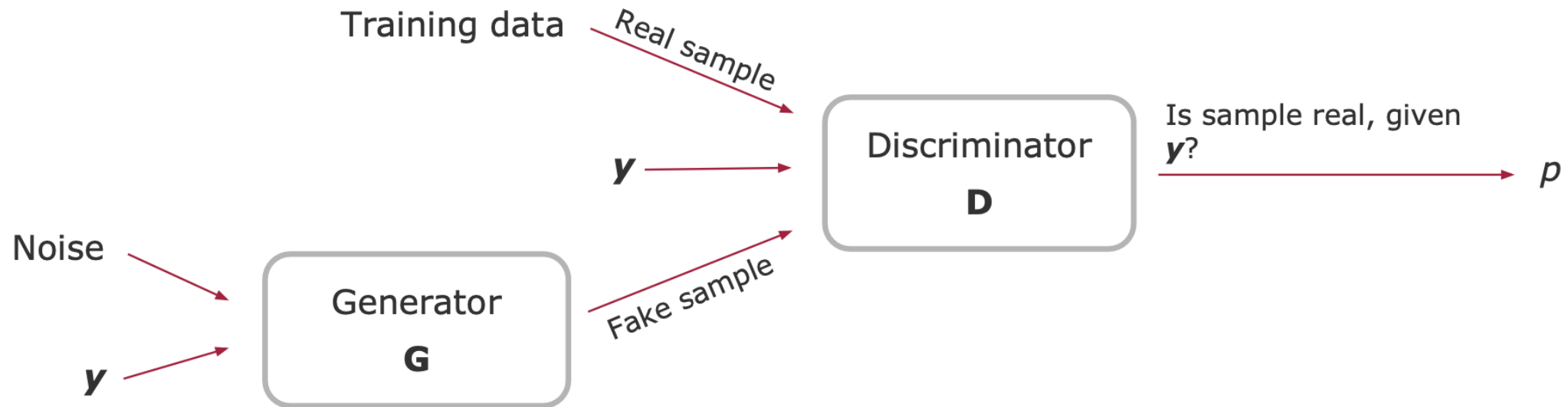
- **Generator (G)** that learns the real data distribution to generate fake samples
- **Discriminator (D)** that attributes a probability p of confidence of a sample being real (*i.e.* coming from the training data)



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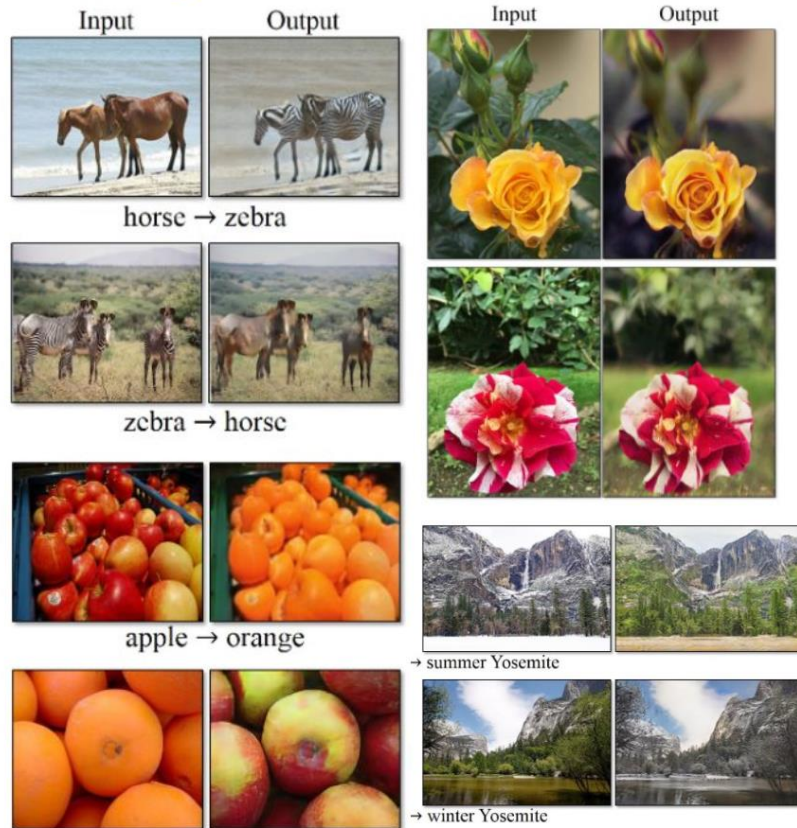
Conditional Generation

- G and D can be **conditioned by additional information y**
- Adding y as an input of both networks will condition their outputs
- y can be external information or data from the training set



More GANs

Source->Target domain transfer



CycleGAN. Zhu et al. 2017.

Text -> Image Synthesis

this small bird has a pink breast and crown, and black primaries and secondaries.



this magnificent fellow is almost all black with a red crest, and white cheek patch.



Reed et al. 2017.



Pix2pix. Isola 2017. Many examples at <https://phillipi.github.io/pix2pix/>

Generation with “control

- Generator is a “black box”
- Separate “Style” and similar “Factors”
- Face
 - Higher Level attribute (Pose, Identity)
 - Fine attributes (freckles, hair, texture)

Results: Style Based Generator

All images in this video
were produced by our generator,
they are not photographs of real people

Limitations of GANs

1. Training instability

- Good sample generation requires reaching Nash Equilibrium in the game, which might not always happen

2. Mode collapse

- When G is able to fool D by generating similarly looking samples from the same data mode

3. GANs were original made to work only with **real-valued, continuous data** (e.g. images)

- *Slight changes* in **discrete data** (e.g. text) are impractical

How to Evaluate Generated Outputs?

- What makes a good generative model?
 - Each generated sample is indistinguishable from a real sample



- Generated samples should have variety



A Number of Advanced Generative Models

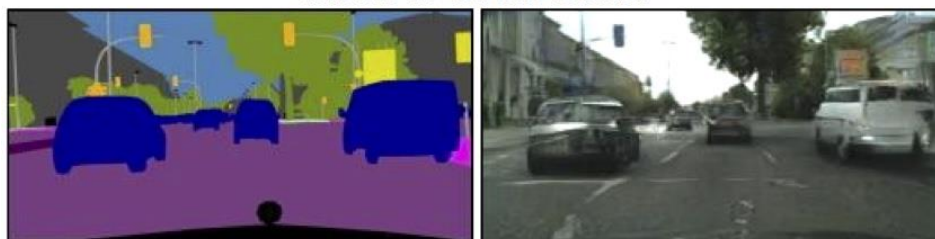
- GANs vs VAEs
- Style GANs and Controllable Generation
- Generation of Multiple Modalities
- Conditional Generation
- Capability to Edit/Process Compact codes
- Diffusion models
- Etc.
- Concern of generation as “fakes”.
 - An emerging ethical concern

Case study

Image to Image Translation

Image to Image Translation with GANs

Labels to Street Scene



input

output

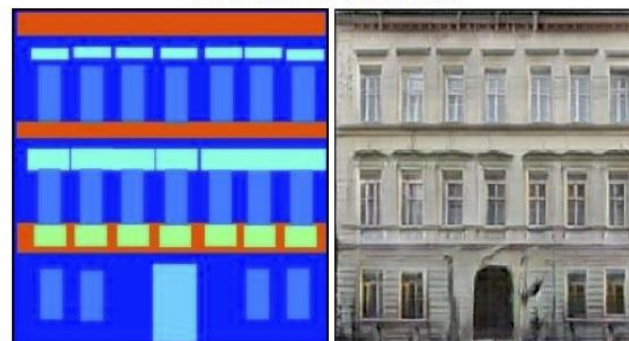
Aerial to Map



input

output

Labels to Facade



input

output

BW to Color



input

output

Day to Night



input

output

Edges to Photo



input

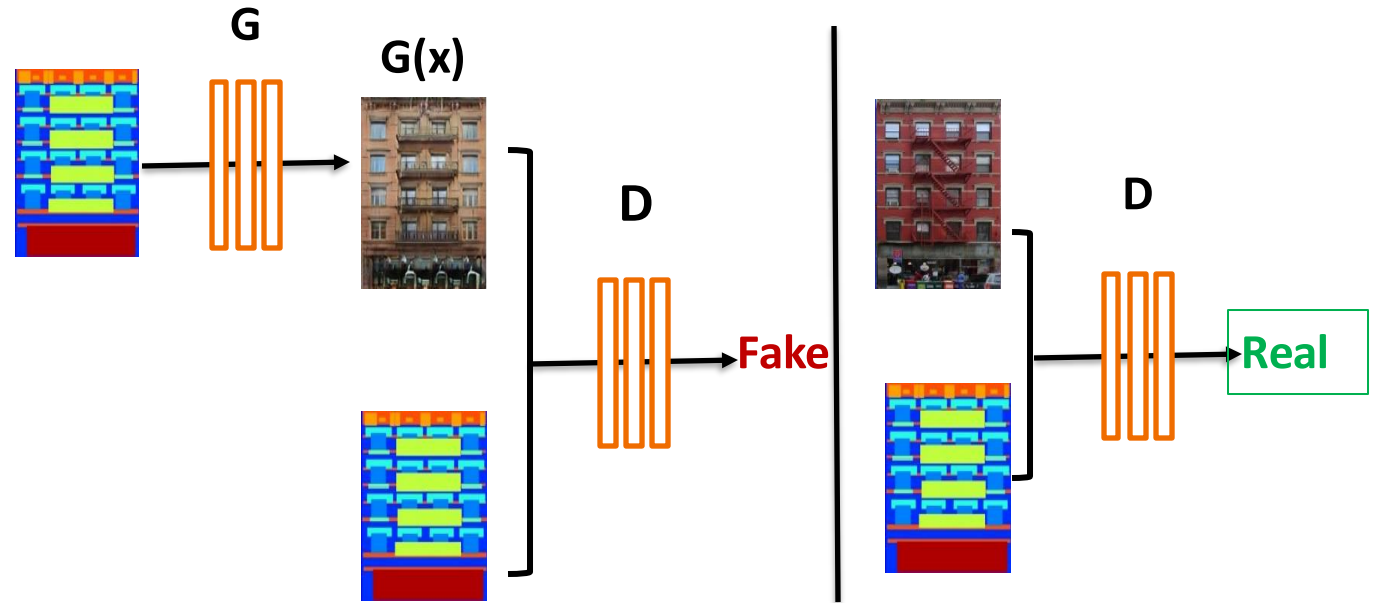
output

Image-to-Image Translation with Conditional Adversarial Networks

Phillip Isola Jun-Yan Zhu Tinghui Zhou Alexei A. Efros

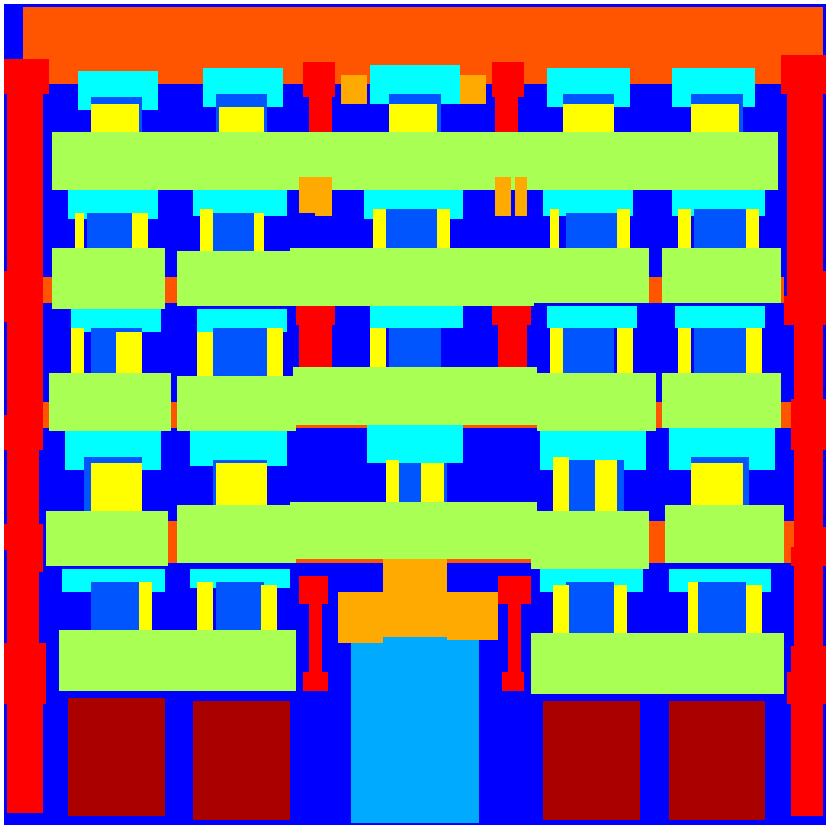
Key Idea

- Training a conditional GAN to map Facade $\rightarrow$ photo
- The discriminator, D , learns to classify between fake and real {edge, photo} tuples
- The generator, G , learns to fool the discriminator
- Unlike an unconditional GAN, both the generator and discriminator observe the input edge map



Results

Input



Output



Data Driven Understanding: Aka Deep Learning



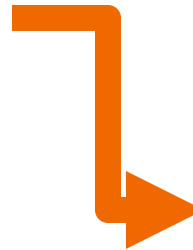
AGE = "40"
Location: (x,y)

Regression

"WALKING" (Action)

Classification

"Male" (Gender)



**Structured
Prediction**

Generative Techniques



Andrej Karpathy, "RNN"

Ian Goodfellow, "GAN"

To prove study we see that $\mathcal{F}|_U$ is a covering of $\mathcal{X}'$, and $\mathcal{T}_i$ is an object of $\mathcal{F}_{X/S}$ for $i > 0$ and $\mathcal{F}_p$ exists and let $\mathcal{F}_i$ be a presheaf of $\mathcal{O}_X$ -modules on $\mathcal{C}$ as a $\mathcal{F}$ -module. In particular $\mathcal{F} = U/\mathcal{F}$ we have to show that

$$\widetilde{M}^\bullet = \mathcal{I}^\bullet \otimes_{\text{Spec}(k)} \mathcal{O}_{S,s} - i_X^{-1} \mathcal{F}$$

is a unique morphism of algebraic stacks. Note that

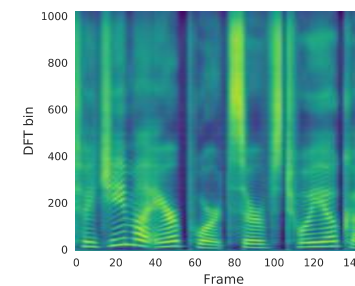
$$\text{Arrows} = (Sch/S)_{fppf}^{opp}, (Sch/S)_{fppf}$$

and

$$V = \Gamma(S, \mathcal{O}) \longmapsto (U, \text{Spec}(A))$$

is an open subset of X . Thus U is affine. This is a continuous map of X is the inverse, the groupoid scheme S .

Proof. See discussion of sheaves of sets. □

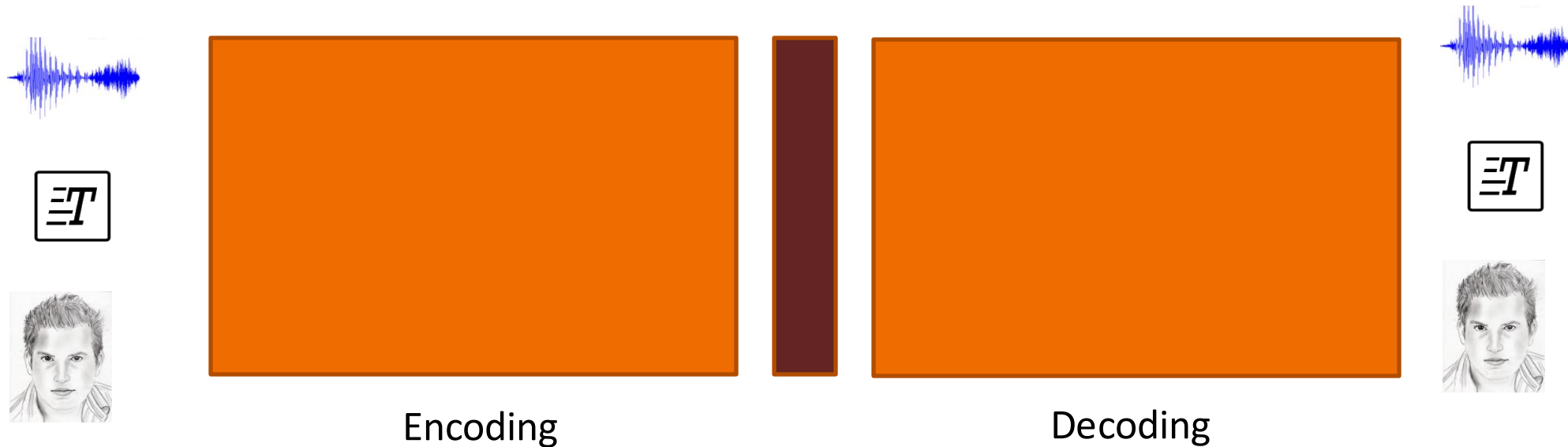


Diffusion



LSTM/Conv1D

Powerful Combination: Translational Models



Inherently multimodal and cross-modal

Summary

- A new training paradigm
 - Adversarial Training
- A set of new methods for generation
 - GANs, VAEs, Diffusion, variants
- Applications
 - Understanding ML, Data, Algorithms
 - Practical Utility vs Ethical Concerns: Race that can never end?
- Cross Modality
 - Transfer and Translation

Thanks!!

Questions?